

UPI Transaction Analysis

Business Context

Unified Payments Interface (UPI) has revolutionized India's digital payments ecosystem, enabling instant, low-cost, and secure money transfers. Your organization—a leading digital payments platform—has rapidly expanded, now managing millions of customers, thousands of merchants, and billions in annual transaction volume.

With this growth come new challenges: fraud risk, compliance requirements, evolving customer behavior, and competitive market pressures.

Key operational questions include:

- Are certain segments at higher risk of transaction fraud?
- Where do bottlenecks or failures occur in the customer or merchant journey?
- How effective are risk controls and fraud alerts?
- How can business performance (by customer, merchant, device, or region) be monitored and improved?

Management seeks to leverage advanced analytics—using SQL, Python, and Power BI—to drive data-driven decision making, optimize operations, enhance security, and deliver value to all stakeholders.

Project Objective

As a Data Analytics Consultant for the Analytics Center of Excellence, you will:

- Analyze operational, transactional, and risk data from UPI systems
- Identify patterns of customer and merchant behavior, payment trends, and fraud risk
- Develop KPIs and dashboards for executive and operational decision-making
- Deliver actionable business recommendations to improve platform performance and user trust

Suggested Project Scope & Tools

Tool	Applications
Excel	Exploratory summaries, scenario modeling, hypothesis tracking
SQL	Data extraction, joins, aggregations from relational databases
Python	Data wrangling, EDA, anomaly detection, visualization, advanced analysis
Statistics	Probability modeling, hypothesis testing, risk scoring, inferential analytics
Power BI / Tableau	Executive dashboard with KPIs, visual analytics, and filters

Project Deliverables

1. Business Understanding & KPI Framework

- Translate business needs into analytical goals
- Define and document KPIs: transaction volume, failure rate, fraud detection rate, customer retention, merchant activity, etc.
- Segment challenges by region, channel, device, and customer type

2. Data Acquisition & Validation

- Import, merge, and validate data using SQL/Python
- Ensure integrity across all tables: customers, devices, UPI accounts, merchants, transactions, feedback, fraud alerts
- Address missing, inconsistent, or anomalous records

3. Exploratory Data Analysis & Descriptive Statistics

- Generate distribution and trend reports: transaction values, merchant activity, device usage
- Identify high-activity and high-risk segments

4. Anomaly and Fraud Detection Analysis

- Analyze patterns associated with fraud alerts
- Model risk drivers (e.g., rooted devices, high-value transactions, merchant risk)
- Compare fraud patterns by region, device, or channel

5. Performance Dashboarding (Power BI)

- Design interactive dashboards for transaction, fraud, and operational KPIs
- Filters by time, region, device type, customer/merchant segment

6. Strategic Insights & Recommendations

- Executive summary of findings (with supporting visualizations)
- Prioritized recommendations for operational improvement and risk management

List of Tables

Table	Description
customer_master	Customer profiles (demographics, join date, etc.)
device_info	Device details and activity (type, rooted status)
upi_account_details	UPI handle data (bank, account type, status)
merchant_info	Merchant master (category, risk, onboard date)
upi_transaction_history	All payment transactions (amount, channel, device, fraud flag)
customer_feedback_surveys	Feedback and satisfaction data (text, score)
fraud_alert_history	Tracked fraud events and resolution

Data Tables Provided

- Customer_Master

Column Name	Data Type	Description
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customer_id	String	Unique customer identifier (e.g., CUST12345)
full_name	String	Customer's full name
age	Integer	Age of the customer
gender	Categorical	Gender (Male/Female/Other)
region	Categorical	Customer's region (North, South, etc.)
date_joined	Date	Date customer joined platform
is_business_user	Boolean	Is customer a business user? (True/False)
risk_score	Float	Customer risk score (0–1, higher = riskier)
mobile_number	String	10-digit mobile number

- Device_Info

Column Name	Data Type	Description
device_id	String	Unique device identifier (e.g., DEV12345)
customer_id	String	FK → customer_master.customer_id
device_type	Categorical	Device type (Android/iOS/Tablet/FeaturePhone)
app_version	String	App version on device
is_rooted	Boolean	Whether device is rooted/jailbroken
last_active	DateTime	Timestamp of last activity on device

- UPI Account Details

Column Name	Data Type	Description
upi_id	String	Unique UPI handle (e.g., anuj1234@upi)
customer_id	String	FK → customer_master.customer_id
bank_name	String	Name of bank associated with UPI ID
account_type	Categorical	Savings/Current/Credit Card Linked
date_added	Date	Date UPI account added
status	Categorical	Status (Active/Blocked/Suspended)

- Merchant Info

Column Name	Data Type	Description
merchant_id	String	Unique merchant identifier (e.g., MERCH1234)
merchant_name	String	Merchant name/company
merchant_type	Categorical	Business category (Grocery, Food, etc.)
region	Categorical	Merchant region (North, South, etc.)
onboard_date	Date	Date merchant onboarded
risk_score	Float	Merchant risk score (0–1)

- UPI Transaction History

Column Name	Data Type	Description
transaction_id	String	Unique transaction identifier (e.g., TXN10000001)
upi_id	String	FK → upi_account_details.upi_id
customer_id	String	FK → customer_master.customer_id
timestamp	DateTime	Date/time of transaction
amount	Float	Transaction amount (INR)
transaction_type	Categorical	Send/Receive/Merchant_Payment/Bill_Pay
merchant_id	String	FK → merchant_info.merchant_id (nullable)
counterparty_upi	String	UPI handle of counterparty (not always in master)
status	Categorical	Status (Success/Failed/Pending)
device_id	String	FK → device_info.device_id
device_type	Categorical	Device type used for transaction
channel	Categorical	Channel (App, QR Code, Intent, etc.)
fraud_flag	Boolean	Transaction flagged as fraud? (True/False)
reversal_flag	Boolean	Transaction reversed? (True/False)
failure_reason	String	Reason for failure (if status=Failed)

- Customer Feedback Surveys

Column Name	Data Type	Description
feedback_id	String	Unique feedback record
customer_id	String	FK → customer_master.customer_id
date_submitted	Date	Date of feedback submission
feedback_text	String	Free-text feedback/comment
satisfaction_score	Integer	Satisfaction rating (1=low, 5=high)
issue_type	Categorical	Type of issue (Transaction, Usability, etc.)
resolved	Boolean	Whether the issue was resolved

- Fraud Alert History

Column Name	Data Type	Description
alert_id	String	Unique fraud alert identifier
transaction_id	String	FK → upi_transaction_history.transaction_id
alert_type	Categorical	Type of fraud alert (Suspicious Login, etc.)
alert_date	DateTime	Date/time of fraud alert
resolved	Boolean	Whether alert was resolved
resolution_date	DateTime	Date/time alert was resolved (nullable)

remarks	String	Additional remarks/comments
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SUGGESTED INSTRUCTIONS

Learners are encouraged to apply their own ideas while executing the project. The following is intended solely as a guideline.

Note: Learners are supposed to check data integrity and handle any inconsistencies accordingly.

Hint: Feedback or certain transaction data may occasionally be missing a customer or merchant reference (for privacy or system reasons).

Step 1: Business Understanding & Analytical Planning

Objective: Develop a clear understanding of the business context and define analytical goals.

- Start with a written note (preferably PowerPoint/Google Slides) outlining:
 - The core business challenges facing your digital payments/UPI platform (e.g., fraud risk, failed transactions, customer churn, merchant engagement, compliance).
 - Key analytical goals from a business intelligence perspective.
 - Identify core KPIs: Transaction Volume, Transaction Failure Rate, Fraud Detection Rate, Customer Retention, Merchant Activity, Device Risk, etc. (add others as needed).
 - Formulate hypotheses related to these KPIs (e.g., "Rooted devices are associated with higher fraud rates" or "Merchant transactions in Metro regions have lower failure rates").
- Present this summary as a brief write-up in PowerPoint/Google Slides.

Step 2: Data Validation Using Excel

Objective: Validate data integrity and address basic inconsistencies before ingestion.

- Verify foreign key relationships. For example:
 - Ensure all customer_id values in upi_account_details, device_info, upi_transaction_history, and customer_feedback_surveys exist in customer_master.
 - Check that all merchant_id in upi_transaction_history exist in merchant_info.
 - Validate that all upi_id in upi_transaction_history exist in upi_account_details.
 - Ensure all device_id in upi_transaction_history exist in device_info.
 - All transaction_id in fraud_alert_history must exist in upi_transaction_history.
- Use Excel's conditional formatting to highlight:
 - Missing or blank fields in critical columns such as amount, status, fraud_flag, risk_score, device_type, etc.
 - Unexpected or inconsistent categorical entries (e.g., typos in status like "succes" instead of "Success", or device types).
- Perform basic data cleaning in Excel, if required:
 - Standardize categorical values (e.g., correcting "Androd" to "Android").
 - Format date columns consistently across all tables.
 - Flag missing values that require imputation or further review (avoid deleting at this stage).
- Create a flag column such as validation_status to tag rows needing review.
- Create a "Data Quality Log" worksheet to summarize:
 - Column names with detected issues

- Type of inconsistency (e.g., missing, invalid, outlier)
 - Suggested cleaning action
 - Count or percentage of affected rows
- Summarize your findings and validation steps in PowerPoint/Google Slides.

Step 3: Database Design and Table Creation (SQL DDL)

Objective: Design and structure the relational database schema.

- Write SQL DDL scripts to create tables for all UPI datasets.
 - Define appropriate data types, primary keys, and foreign key constraints based on the provided data dictionary.
 - Include NOT NULL, UNIQUE, and CHECK constraints as applicable.
- Save all DDL scripts in a text file and validate their execution in your SQL environment.
- Optionally, draw an ER diagram showing table relationships.

Step 4: Data Ingestion into SQL

Objective: Import cleaned data into a structured relational database.

- Load the Excel or CSV datasets into their respective SQL tables.
 - Use native import features or SQL queries like LOAD DATA INFILE, or tools like MySQL Workbench.
 - Optionally, automate using Python (pandas + SQLAlchemy).
- After ingestion, validate:
 - Row counts across tables

- Random spot checks for correct field mapping
- Consistency of foreign key relationships after loading

Step 5: Connecting Python to SQL

Objective: Establish a dynamic data pipeline from SQL to Python for analysis.

- In a Jupyter Notebook, demonstrate:
 - Installation/import of required libraries (pymysql, sqlalchemy, pandas)
 - Construction of a secure connection string to your SQL database
 - Executing SQL queries from Python using `pandas.read_sql_query`
 - Loading data from all tables into Pandas dataframes for further use
 - Validating schema and record counts in Python to ensure successful extraction

Step 6: Data Cleaning, Transformations & Enhanced EDA in Python

Objective: Prepare data for analysis and uncover initial trends.

- Clean the extracted data:
 - Treat missing values through imputation or removal as justified.
 - Normalize and standardize key numerical fields (e.g., `risk_score`, `amount`). Interpret and document any transformations.
 - Create derived columns (examples only):
 - $\text{Transaction Value per Customer} = \frac{\text{Total Transaction Amount}}{\text{Number of Transactions}}$

- $\text{Merchant Fraud Ratio} = \frac{\text{Fraud Transactions}}{\text{Total Transactions by Merchant}}$
- $\text{Device Fraud Ratio} = \frac{\text{Fraud Transactions}}{\text{Total Transactions by Device}}$
- $\text{Transaction Failure Rate} = \frac{\text{Failed Transactions}}{\text{Total Transactions}}$
- Perform EDA using pandas, matplotlib, and seaborn:
 - Visualize trends over time (e.g., daily/monthly transaction/fraud volumes)
 - Use heatmaps and boxplots to detect regional, merchant, and device-level patterns
 - Summarize metrics using grouped statistics (mean, median, standard deviation)
 - Identify high-activity and high-risk segments

Step 7: Statistical Analysis in Python

Objective: Validate business hypotheses using statistical tests.

- Perform statistical tests using scipy.stats and statsmodels. Examples:
 - T-tests to compare average transaction amounts across device types or regions
 - ANOVA to evaluate variability in fraud rates by merchant type or region
 - Chi-square test for associations between fraud flag and channel, or between transaction status and device type
 - Correlation analysis between customer risk_score and transaction/fraud occurrence
- Interpret test outputs, including p-values, confidence intervals, and appropriate visualizations (boxplots, histograms, etc.).
- Document assumptions and limitations of the tests performed.

Step 8: Power BI / Tableau Dashboard Development

Objective: Build executive dashboards for performance monitoring and decision support.

- Connect Power BI Desktop or Tableau to your SQL database.
- Build an integrated data model using all relevant tables.
- Create the following dashboards (examples):
 - **Executive Dashboard:**
KPIs: Total Transactions, Avg Transaction Amount, Failure Rate, Fraud Rate, Top Devices/Merchants
Visuals: Line charts for trends, donut charts for transaction type, bar graphs for region/device/merchant comparison
Filters: Region, Device Type, Merchant, Status, Time Period
 - **Operations/Fraud Analyst Dashboard:**
Drill-down views for fraud alert trends, high-risk devices/merchants, root cause analytics
- Ensure dashboards are clean, interactive, and easy to interpret for non-technical audiences.

Step 9: Strategic Insights & Recommendations

Objective: Derive actionable business insights and communicate them effectively.

- Draft a concise 2-page executive report covering:
 - Top 5 data-driven insights (e.g., segments/devices with high fraud rates, underperforming merchant categories, root causes of failed transactions)
 - Supporting visualizations from Power BI/Tableau or Python
 - At least 3 strategic recommendations (e.g., targeted device security measures, merchant onboarding reviews, customer education campaigns)

- Ensure recommendations are backed by metrics and include potential business impact estimates